Case Assignment: Book Transactions

1. Data Restructuring:

Step 1: Split the data into a calibration sample comprising all records on or before 9-30-17, and a validation sample comprising all records after 9-30-17.

Calibration sample size: 49086
Validation sample size: 20573

Step 2: For both samples restructure the data so that there is one row per customer with the following columns: “ID”, “frequency” (number of purchases), “monetary” (average dollar value of purchases), “last.purchase” (the date of the customer’s most recent purchase)

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<class 'pandas.core.frame.DataFrame'>
RangeIndex: 69659 entries, 0 to 69658
Data columns (total 4 columns):
 #   Column   Non-Null Count  Dtype
---  ------   --------------  -----
 0   ID       69659 non-null  int64
 1   BOOKS    69659 non-null  int64
 2   DOLLARS  69659 non-null  float64
 3   DATE     69659 non-null  datetime64[ns]
```

Step 3: For both samples, create a column called “recency” which is a numeric variable that represents how many days since the most recent purchase. A good way to do this is to use the last date in the whole sample as an end date. So, a purchase on that date would have a recency score of 0.

calibration_rfm

	ID	monetary	frequency	lastpurchase	recency
0	1	11.770000	1	2017-01-01	272
1	2	44.500000	2	2017-01-12	261
2	3	20.353333	3	2017-04-02	181
3	4	24.673333	3	2017-08-02	59
4	5	32.651250	8	2017-09-15	15

validation_rfm

	ID	monetary	frequency	lastpurchase	recency
0	3	31.800000	3	2018-05-28	33
1	4	26.480000	1	2017-12-12	200
2	5	41.466667	3	2018-01-03	178
3	7	117.965000	2	2018-03-22	100
4	8	22.967500	4	2018-03-29	93

Step 4: Merge the calibration and validation samples into one data frame with one row per customer ID.

ID	monetary_calibration	frequency_calibration	lastpurchase_calibration	recency_calibration	monetary_validation	frequency_validation	lastpurchase_validation	recency_validation
1	11.770000	1	2017-01-01	272	NaN	NaN	NaT	NaN
2	44.500000	2	2017-01-12	261	NaN	NaN	NaT	NaN
3	20.353333	3	2017-04-02	181	31.800000	3.0	2018-05-28	33.0
4	24.673333	3	2017-08-02	59	26.480000	1.0	2017-12-12	200.0
5	32.651250	8	2017-09-15	15	41.466667	3.0	2018-01-03	178.0

Step 5: Create a column that takes value 1 if the customer was retained (i.e. if the customer made at least one purchase in the second period) and a 0 if not.

Now sort your new data by ID, and print out the first 10 rows of your newly constructed data in your report.

	ID	monetary_calibration	frequency_calibration	lastpurchase_calibration	recency_calibration	monetary_validation	frequency_validation	lastpurchase_validation	recency_validation	retained
0	1	11.770000	1	2017-01-01	272	NaN	NaN	NaT	NaN	0
1	2	44.500000	2	2017-01-12	261	NaN	NaN	NaT	NaN	0
2	3	20.353333	3	2017-04-02	181	31.800000	3.0	2018-05-28	33.0	1
3	4	24.673333	3	2017-08-02	59	26.480000	1.0	2017-12-12	200.0	1
4	5	32.651250	8	2017-09-15	15	41.466667	3.0	2018-01-03	178.0	1
5	6	20.990000	1	2017-01-01	272	NaN	NaN	NaT	NaN	0
6	7	28.740000	1	2017-01-01	272	117.965000	2.0	2018-03-22	100.0	1
7	8	26.447500	4	2017-07-03	89	22.967500	4.0	2018-03-29	93.0	1
8	9	26.935000	2	2017-05-13	140	41.980000	1.0	2018-06-08	22.0	1
9	10	39.310000	1	2017-01-21	252	NaN	NaN	NaT	NaN	0

2. Predicting retention with Logistic regression: predict retention in period 2 using RFM in period 1. Plot out the ROC curve for the logistic regression model. What is the AUC (area under the curve)?

```
Optimization terminated successfully.
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         Iterations 6
                           Logit Regression Results
==============================================================================
Dep. Variable:               retained   No. Observations:                23570
Model:                          Logit   Df Residuals:                    23566
Method:                           MLE   Df Model:                            3
Date:                Thu, 13 Mar 2025   Pseudo R-squ.:                  0.1909
Time:                        00:46:58   Log-Likelihood:                -11640.
converged:                       True   LL-Null:                       -14387.
Covariance Type:            nonrobust   LLR p-value:                     0.000
=========================================================================================
                            coef    std err          z      P>|z|      [0.025      0.975]
-----------------------------------------------------------------------------------------
Intercept                 0.2037      0.073      2.804      0.005       0.061       0.346
recency_calibration      -0.0096      0.000    -34.436      0.000      -0.010      -0.009
frequency_calibration     0.2449      0.014     18.004      0.000       0.218       0.272
monetary_calibration      0.0022      0.001      4.146      0.000       0.001       0.003
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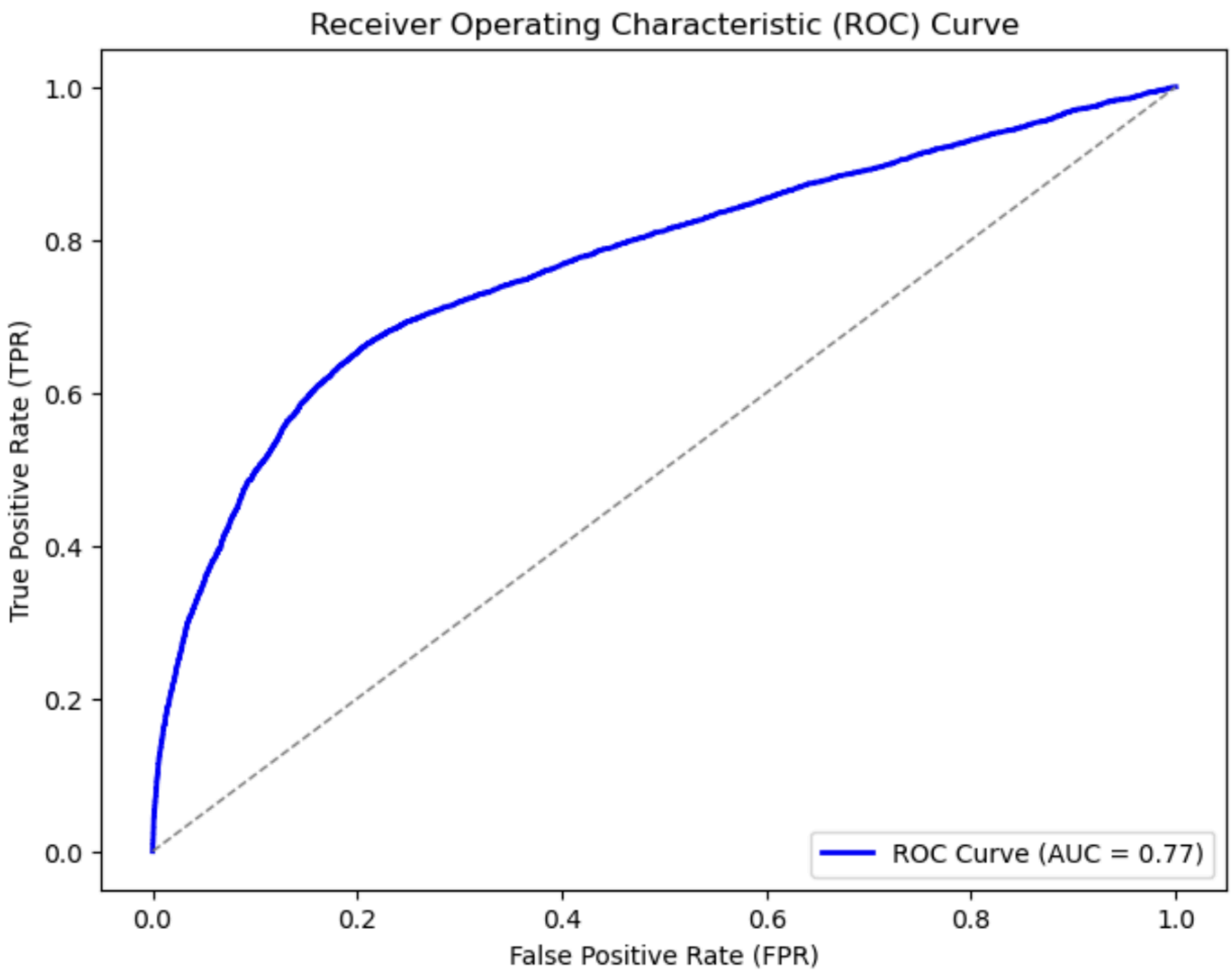


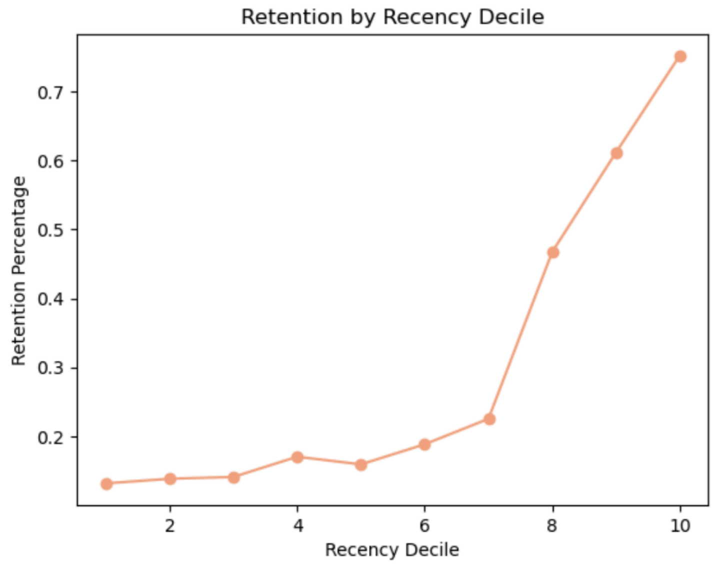
From our AUC of **0.77**, we can say that the model is moderately good at distinguishing between the positive(those who stay) and negative(those who leave) classes. (as an AUC of 0.5 means no better than random guessing, while 1.0 would be perfect.)

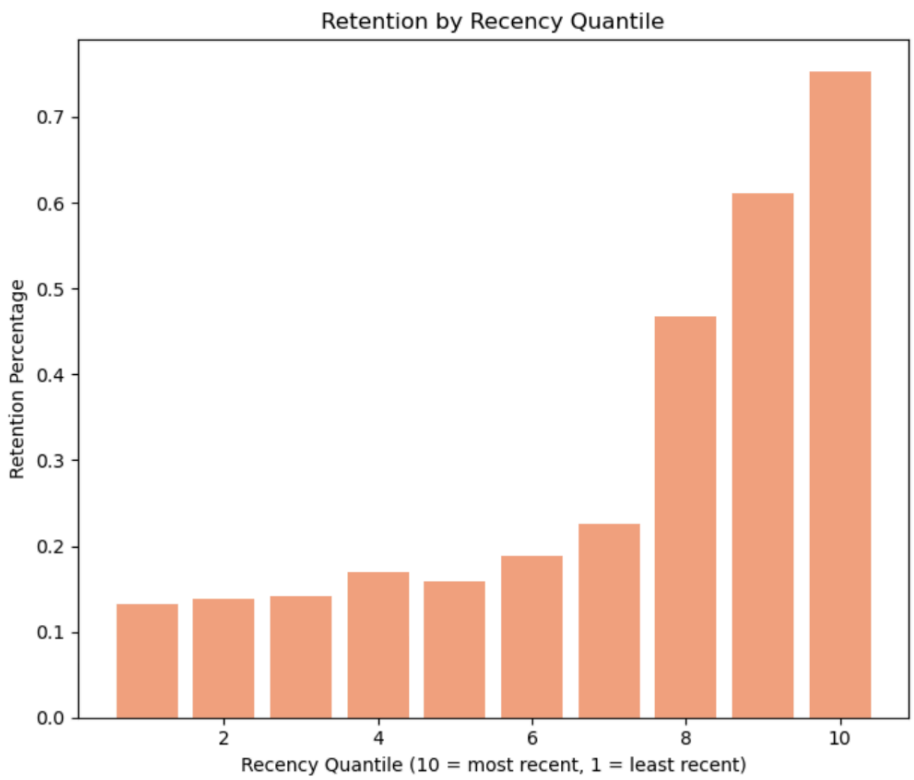
3. Decile analysis: For recency.x, frequency.x, and monetary.x in the restructured data, create new columns that place each customer into a decile based on how they rank. Calculate and plot the percent retention in each decile of recency, each decile of frequency, and each decile of monetary. Do these variables seem to make any difference to retention?

Recency

Recency Quantile Summary:

	recency_quantile	retained_number	count	retention_percentage
0	10	1801	2394	0.752297
1	9	1454	2377	0.611695
2	8	1076	2302	0.467420
3	7	580	2576	0.225155
4	6	408	2168	0.188192
5	5	382	2402	0.159034
6	4	393	2310	0.170130
7	3	331	2351	0.140791
8	2	339	2454	0.138142
9	1	294	2236	0.131485

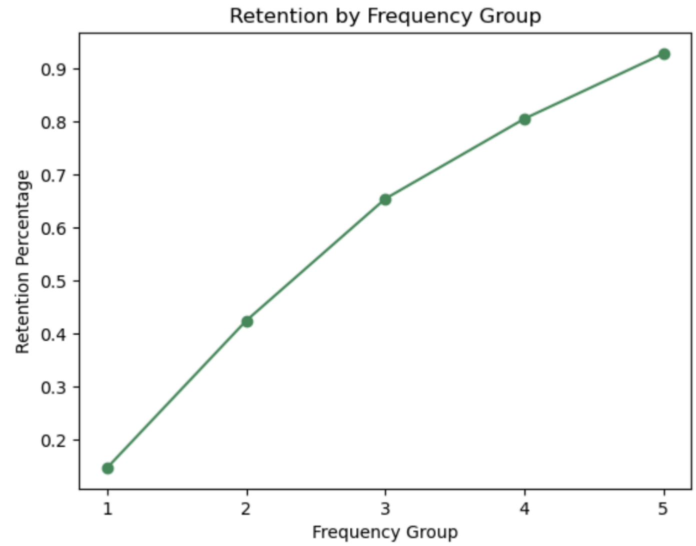


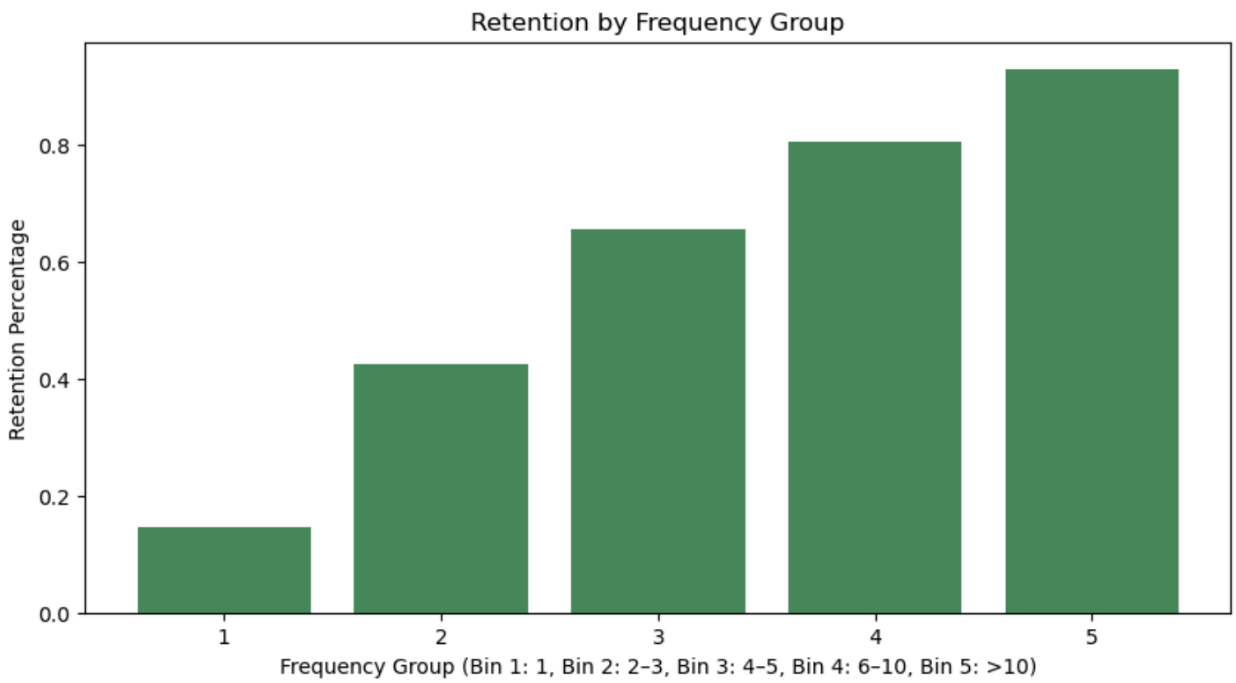


Insights- Customers who engaged more recently (higher recency quantiles) have a much higher retention rate ie. around 75% for the most recent group, compared to about 13% for the least recent group. This strong upward trend indicates that recent engagement is a key predictor of retention, suggesting that re-engagement efforts for less recent customers could significantly improve overall retention.

Frequency:

	frequency_group	retained_number	count	retention_percentage
0	1	2046	13954	0.146625
1	2	2793	6585	0.424146
2	3	1114	1703	0.654140
3	4	834	1036	0.805019
4	5	271	292	0.928082

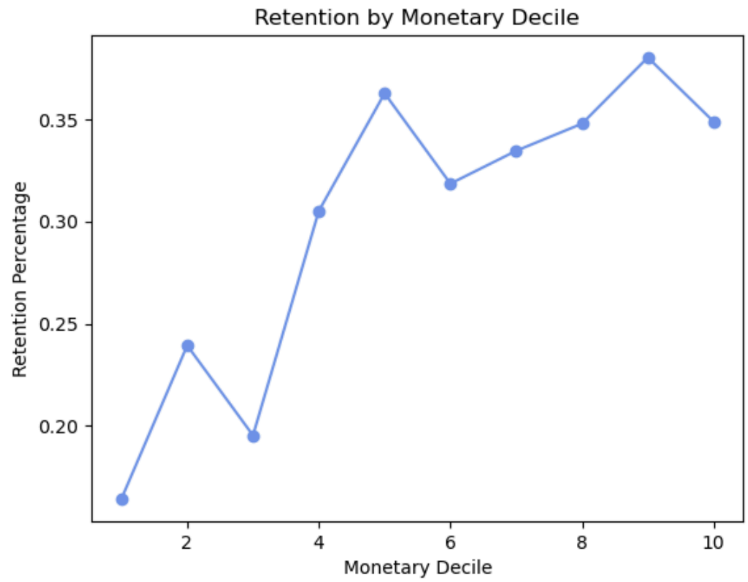


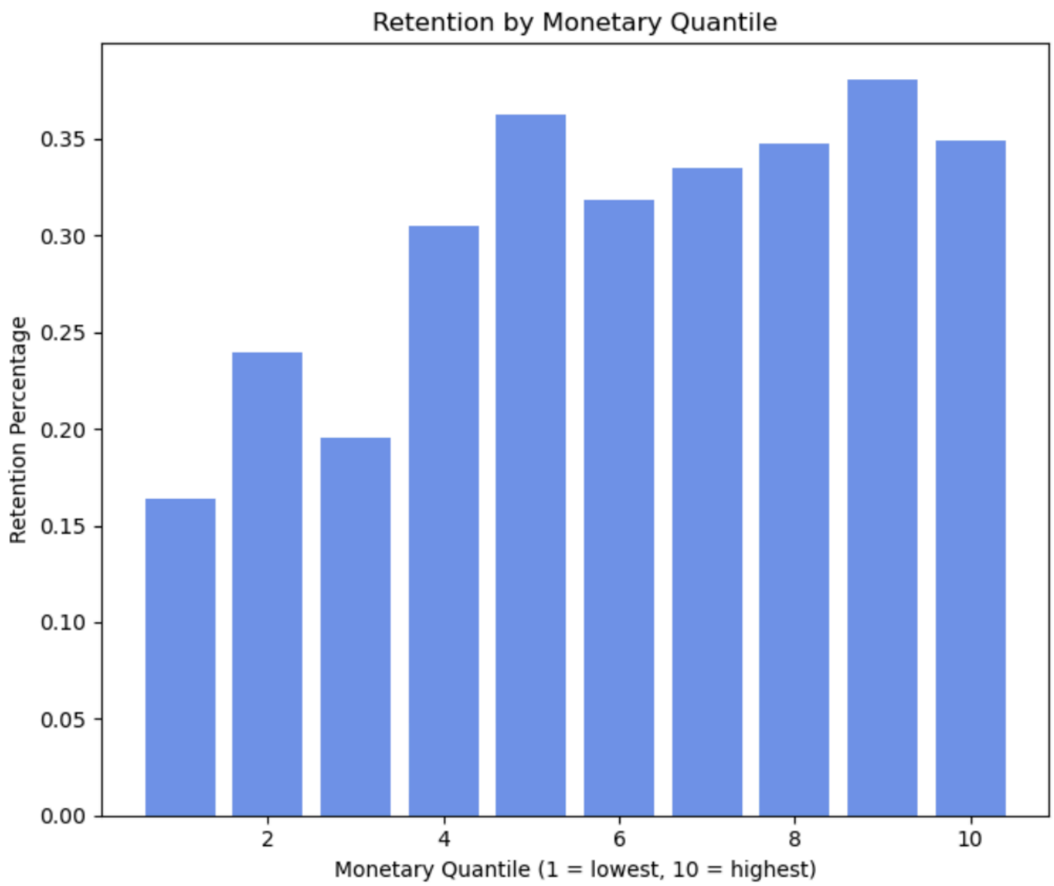


Insights- Customers who purchase more frequently have substantially higher retention rates. Retention improves from around 15% for single-purchase customers to over 90% for those with more than 10 purchases. From this upward trend we can say that frequency is a strong predictor of retention, and we should encourage repeat purchases for improving overall customer retention.

Monetary:

	monetary_quantile	retained_number	count	retention_percentage
0	1	387	2357	0.164192
1	2	564	2357	0.239287
2	3	470	2407	0.195264
3	4	704	2307	0.305158
4	5	855	2357	0.362749
5	6	753	2364	0.318528
6	7	787	2352	0.334609
7	8	820	2357	0.347900
8	9	896	2356	0.380306
9	10	822	2356	0.348896



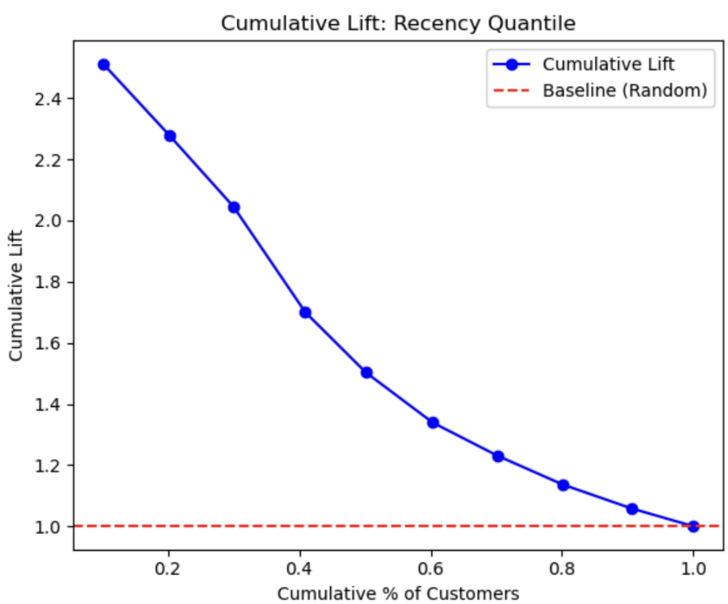


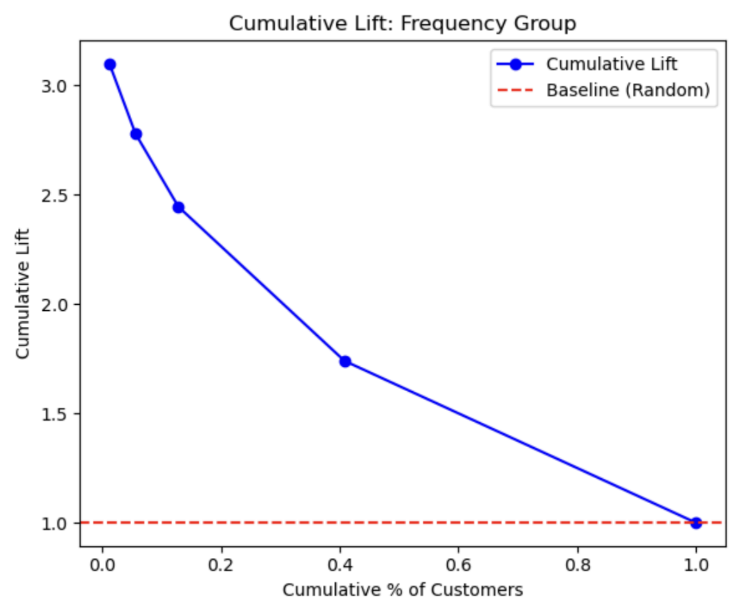
Insights- Customers who spend more (higher monetary deciles) tend to have somewhat higher retention rates, rising from around 16% in the lowest group to roughly 34-38% at the higher deciles. However, the relationship is not perfectly monotonic (there are some fluctuations), suggesting monetary value does influence retention, but less consistently than recency or frequency.

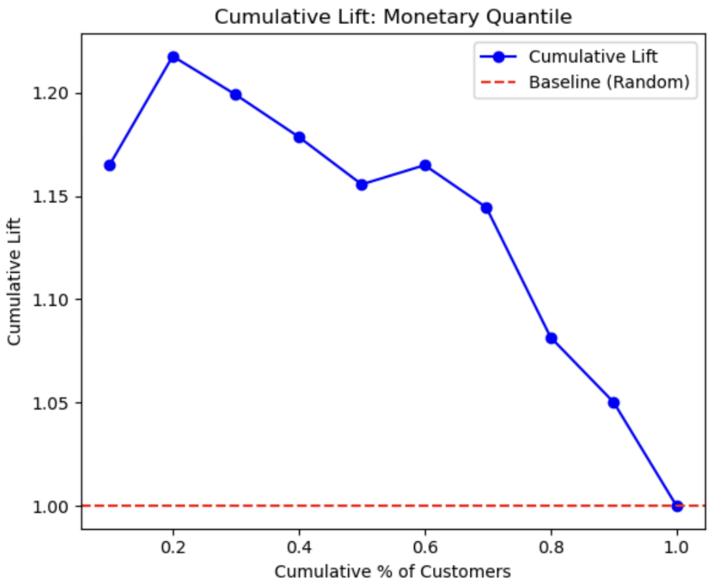
Factor	Observed Effect on Retention	Recommendations
Recency	- Strong upward trend: Customers with the most recent purchases have significantly higher retention. - Retention can jump from around 13% in the least recent group to 75% in the most recent group.	- Implement re-engagement campaigns for customers who have not purchased/visited recently. - Offer targeted promotions or reminders to encourage less-recent customers to return.
Frequency	- Positive relationship: More frequent purchasers show higher retention. - Retention increases from ~15% for single-purchase customers to ~92% for customers with >10 purchases.	- Encourage repeat purchases via loyalty programs, subscription models - Use cross-sell(similar products) or up-sell tactics(more premium product) to boost purchase frequency.
Monetary	- Moderate correlation: higher-spending customers tend to have better retention, but with some fluctuations. - Retention ranges from ~16% in the lowest decile to 34-38% in the higher deciles.	- Tiered rewards based on spend levels (e.g., VIP tiers). - Targeted incentives to convert mid-level spenders into higher-value customers..

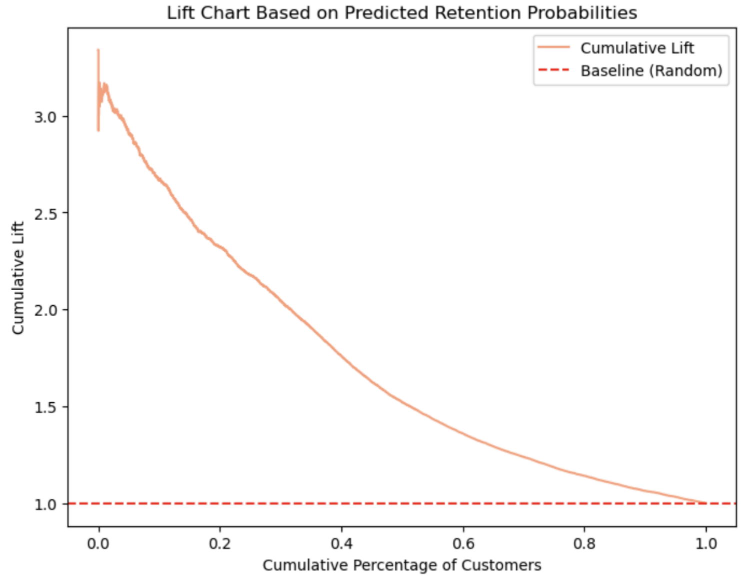
4. Response models are commonly evaluated in terms of their selectivity, or their ability to sort customers according to their response potential. Based on the estimated logistic regression model,

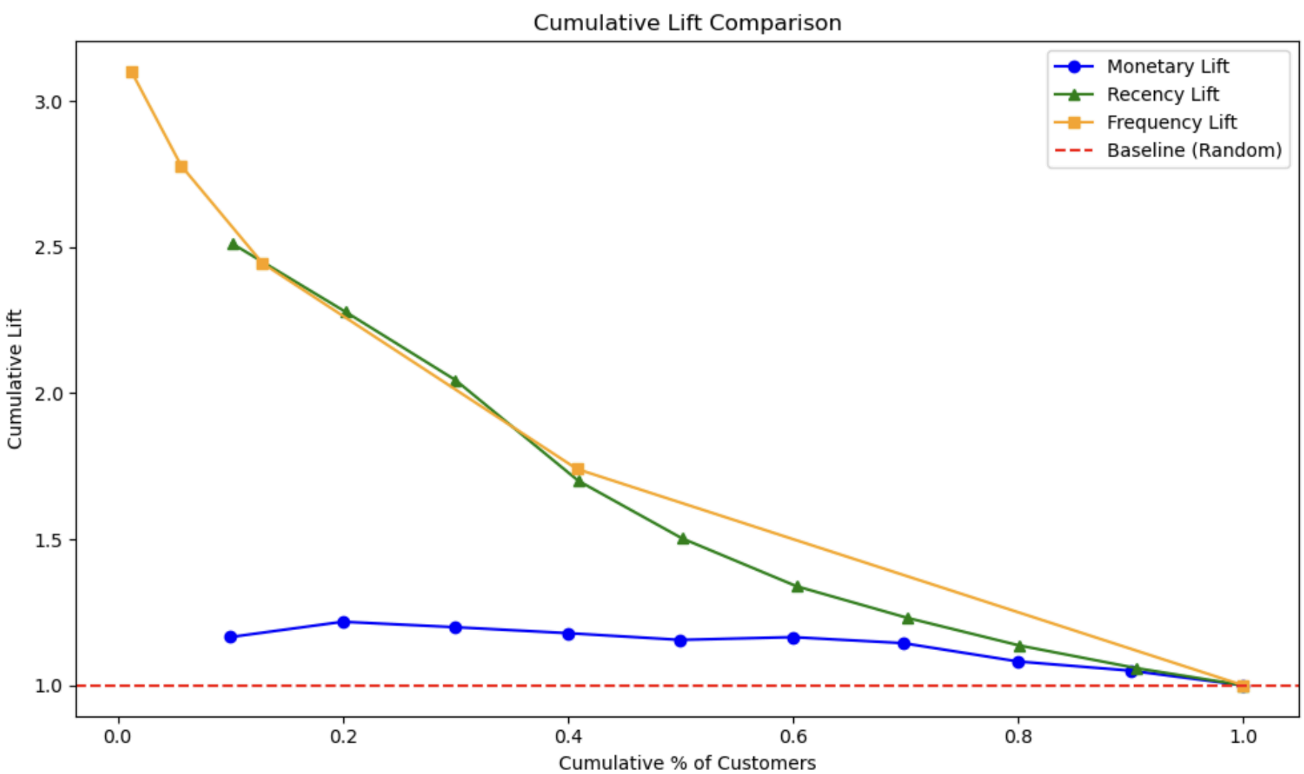
a. Can you assess the customer selectivity ability of the variable “Recency.x” by plotting out the cumulative lift chart? What about the selectivity ability of the variable “Frequency.x”? What about the selectivity ability of the variable “Monetary.x”? What about the selectivity ability using the predicted retention probability of the logistic regression model?











A cumulative lift chart measures how much better a model is in identifying high retention customers, as compared to a random selection (shown by the red line).

1. Recency

Recency has a high initial lift (around 2.5×), reflecting that the most recent buyers are about 2.5 times more likely to be retained compared to the average. The lift gradually declines as more customers are included. Still, recency is a close second to frequency in predictive power.

2. Frequency

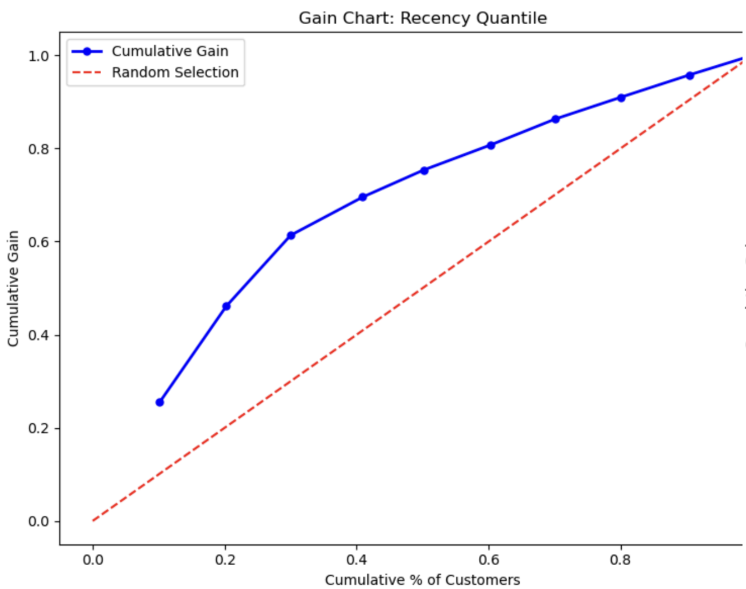
Frequency provides the strongest initial lift (around 3×), this shows that the top segment is retaining at about 3 times the overall average, indicating that top buyers are far more likely to remain. Hence, focusing on frequent purchasers can significantly boost overall retention.

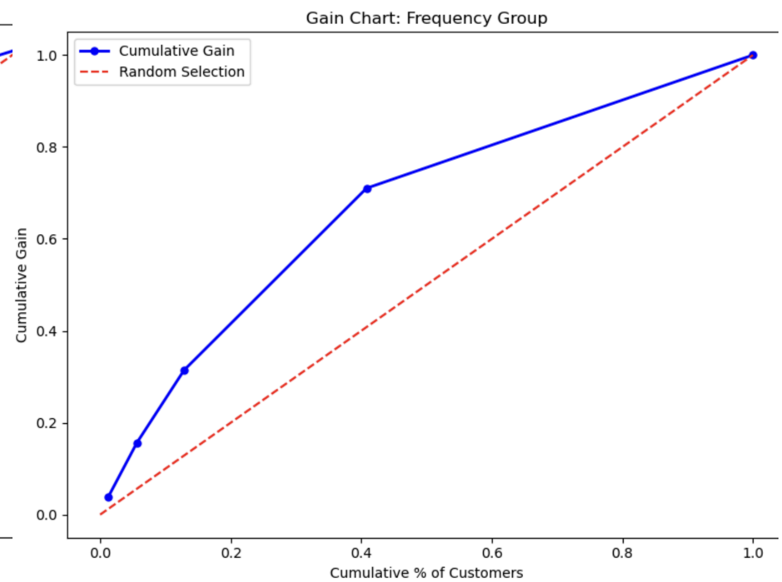
3. Monetary
Monetary offers a lower lift overall (starting around 1.25), so top spenders outperform the average but not as dramatically as frequent or recent buyers. Its selectivity is weaker compared to frequency or recency, but still a relevant factor to consider.

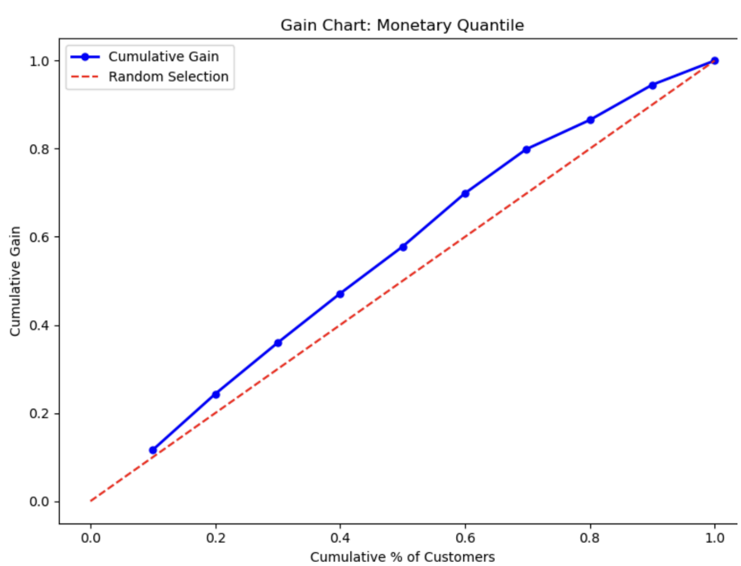
4. Logistic Regression model
The logistic model is achieving a high initial lift (about 3×) by combining multiple variables. It effectively ranks and isolates high-retention customers, outperforming any single factor(close to frequency though). Overall, this multi-factor(RFM) approach is yielding the best predictive performance.

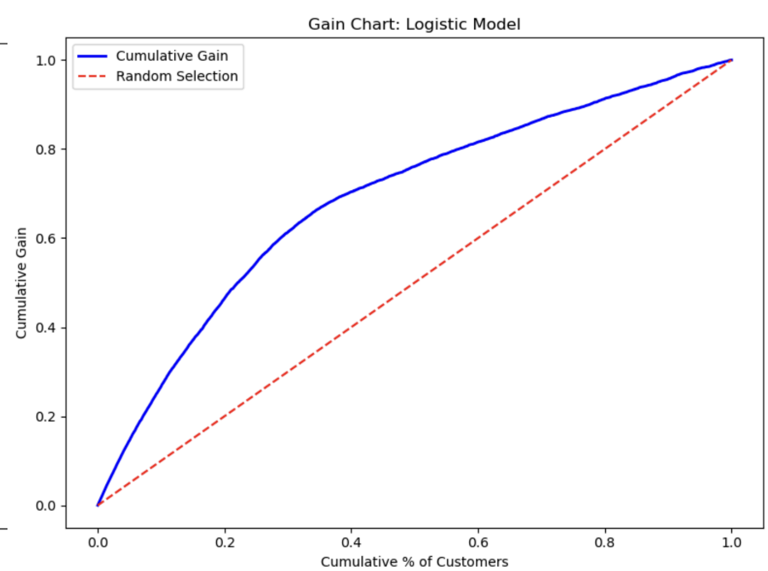
Hence we can conclude that frequency and recency are the strongest predictors of retention, with the top segments outperforming the average by around 3× and 2.5×, respectively. Monetary value, while still relevant, provides less selectivity. The logistic regression model reinforces these insights by achieving high initial lift (around 3×).

b. Can you assess the customer selectivity ability of the variable "Recency.x" by plotting out the gain chart? Plot out the gain chart using variables, "Frequency.x", "Monetary.x", and the predicted retention probability of the logistic regression model, respectively.









The Gain Chart shows how well each variable identifies high-retention customers by measuring the percentage of retained customers captured at different targeting levels

1. Recency Quantile
The gain curve rises rapidly above the baseline, with the top 30% of customers yielding 60% of retained customers. This indicates that recent purchasers are strongly predictive of retention. Focusing on this top segment can efficiently capture a large share of retained customers.
2. Frequency Group
The gain curve climbs steeply, with the top 40% of high-frequency customers capturing 70% of retained customers. This demonstrates that customers who purchase more often are highly likely to be retained. Targeting this segment provides a significant improvement over random selection.
3. Monetary Quantile
The gain curve for monetary quantiles is moderately above the baseline, with the top 70% yielding 80% of retained customers. This shows that high-spending customers are retained better than average, but the separation is less pronounced. Monetary value thus offers moderate selectivity compared to recency and frequency.
4. Logistic Model
The logistic model's gain curve remains well above the baseline across segments, reflecting strong overall selectivity. By integrating all 3 RFM features, it is accurately ranking customers by retention probability and outperforming any single metric in identifying high-retention customers.

Hence, we can conclude that frequency and recency are the strongest predictors, with their top segments capturing a high proportion of retained customers.Monetary still provides some gain, but its ability to identify high-retention customers is comparatively weaker. The logistic model, by integrating all these variables, achieves the highest overall gain, confirming that a multi-factor approach outperforms reliance on any single metric.